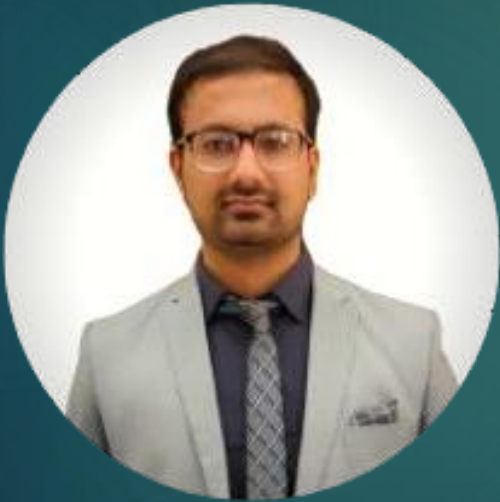


XML Introduction



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XML ?

eXtensible Markup Language

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XML ?

- It is used for storing and transferring **data**.
- It is a markup language much like HTML.
- Unlike HTML, XML is **case-sensitive**, requires each tag is closed properly, and preserves whitespace.
- XML Tags need not to be **predefined** like HTML (known as extensible).

XML ?

- A markup language is a language that **annotates** text so that the computer can **manipulate** that text.
- Most markup languages are **human-readable** because the annotations are written in a way to distinguish them from the text itself.
- It was designed to be **self-descriptive**.

XML in Android

- We create XML **layouts** in Android, and later alter them using Java logic.
- **Resources** are the additional files and static content an application needs, such as animations, color schemes, layouts, menu layouts.
- Each layout file must contain one (and only one!) **Root** element.

View

- It represents a rectangular area of the screen, and is responsible for **displaying information or content**, and event handling.
- Text, images, and buttons are all Views in Android.

ViewGroup

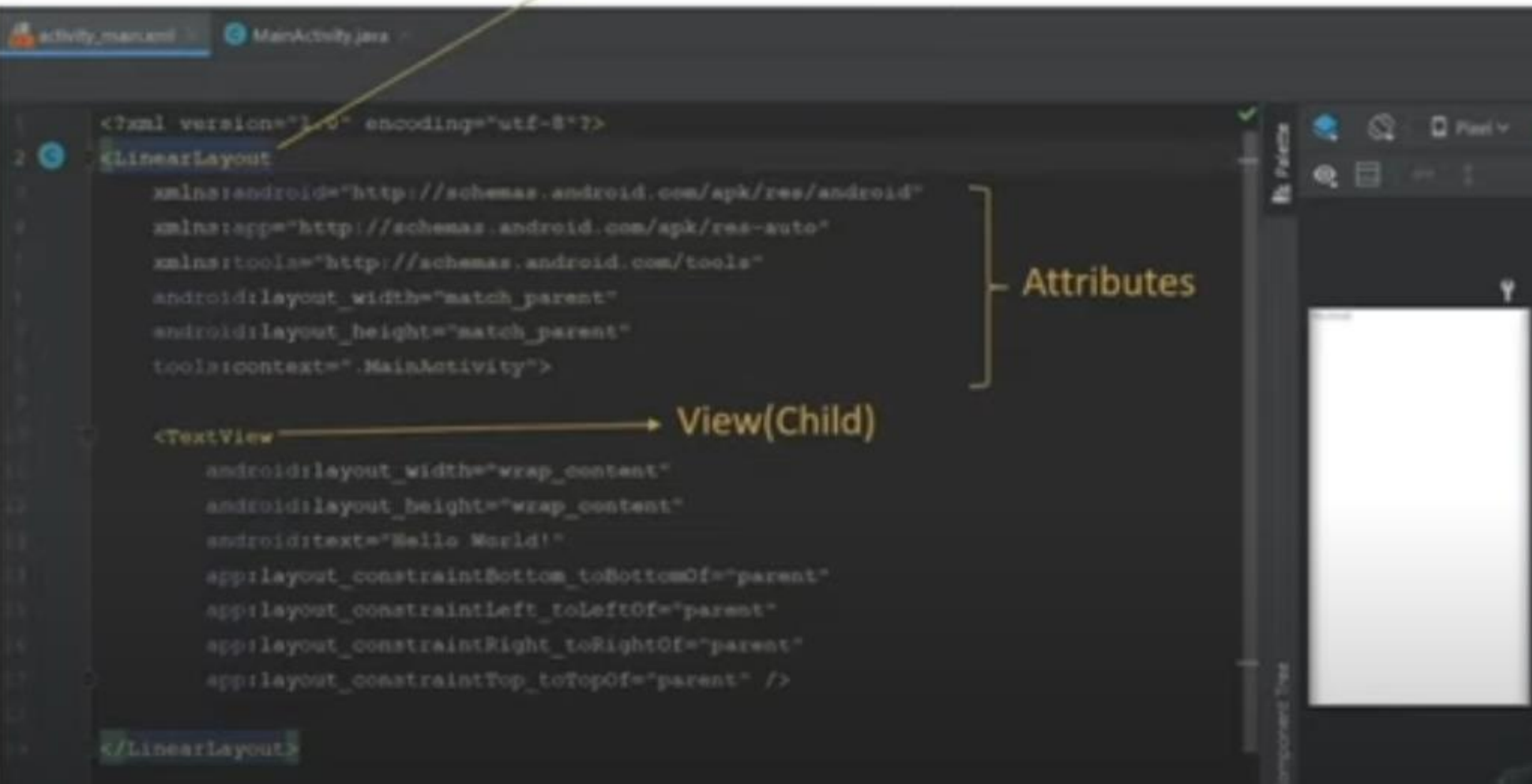
- It is essentially an '**invisible container**' that holds multiple Views or ViewGroups together, and defines their layout properties.
- Common View Groups
 - A **List View** displays a list of scrollable items.
 - A **Grid View** displays items in a two-dimensional scrollable grid.
 - A **Table Layout** groups views into rows and column

Root View

- It is **Root** Element of XML Layout file.
- Common View Groups
 - A **Linear Layout** aligns its contents into a single direction, whether vertical or horizontal.
 - A **Relative Layout** displays its child content in positions relative to the parent.
 - A **Frame Layout** is a placeholder on a screen that display only a single view (Fragments)

XML Layout

Root View



The screenshot shows the XML layout editor in Android Studio. The XML code is as follows:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</LinearLayout>
```

Annotations in the image:

- Root View**: Points to the opening `<LinearLayout>` tag.
- Attributes**: A bracket on the right side of the `<LinearLayout>` tag groups its attributes: `xmlns:android`, `xmlns:app`, `xmlns:tools`, `android:layout_width`, `android:layout_height`, and `tools:context`.
- View(Child)**: Points to the `<TextView>` tag.

The right sidebar shows the 'Run' button and a preview of the layout, which is a white rectangle.

XML Layout

Root View

MyFirstApp / app / src / main / res / layout / activity_main.xml

activity_main.xml MainActivity.java

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    tools:context=".MainActivity">

    <GridLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:rowCount="3"
        android:columnCount="3">

        <androidx.appcompat.widget.AppCompatButton/>
        <androidx.appcompat.widget.AppCompatButton/>
        <androidx.appcompat.widget.AppCompatButton/>
        <androidx.appcompat.widget.AppCompatButton/>

    </GridLayout>

</LinearLayout>
```

Root View

View Group

Component Tree